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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 06

Lab Report Name : Router Configuration and MAC Filtering for User Access
Control

Course Title : Computer Networks Lab

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Experiment No: 06

Experiment Name: Router Configuration and MAC Filtering for User Access Control

Introduction / Theory

A wireless router provides network connectivity to multiple users and devices. Because wireless signals can be accessed by devices within range, security and access control are important. The router can protect the wireless network using WPA/WPA2 security and a password. In addition, MAC Filtering can be used as an extra access-control mechanism. Every network interface has a unique Media Access Control (MAC) address. By adding a device MAC address to the router's filtering list, the administrator can define whether that device is allowed or denied access to the wireless network.

In this experiment, the router configuration interface was used to check wireless security, identify connected DHCP clients, and configure MAC Filtering. The 5 GHz wireless band was selected for the access-control configuration.

Objectives:

1. To configure the wireless security settings of a TP-Link AC750 wireless dual-band router.
2. To identify connected devices and their MAC addresses from the DHCP Clients List.
3. To understand the purpose and operation of Wireless MAC Filtering.
4. To configure MAC-based access control for a selected wireless device.
5. To enable or disable device access to the wireless network using filtering rules.

Necessary Materials and Software:

1. TP-Link AC750 Wireless Dual Band Router (Archer C20)
2. Wireless client devices
3. Computer or smartphone with a web browser
4. Router web-based administration interface

Procedure:

Step 1: Configure Wireless Security

The Wireless 5 GHz → Wireless Security page was opened. WPA/WPA2-Personal was selected, WPA2-PSK with AES encryption was configured, and the wireless password was set. This protects the wireless network before applying MAC-based access control.

Step 2: Check Connected Devices

The DHCP → DHCP Clients List page was opened to view the devices currently connected to the router. The list displayed the client's name, MAC address, assigned IP address, and lease time. The target device MAC address was identified from this list.

The screenshot shows the TP-Link Archer C20 web interface. The left sidebar contains a menu with options: Status, Quick Setup, Operation Mode, Network, Dual Band Selection, Wireless 2.4GHz, Wireless 5GHz, Guest Network, DHCP (highlighted), - DHCP Settings, - DHCP Clients List, - Address Reservation, Forwarding, Security, Parental Controls, Access Control, Advanced Routing, Bandwidth Control, IP & MAC Binding, Dynamic DNS, IPv6, System Tools, and Logout. The main content area is titled 'DHCP Clients List' and includes a sub-header 'DHCP Clients List'. Below this, a message states: 'This page displays information of all DHCP clients on the network.' A table follows, listing two clients. The first client has ID 1, Client Name 'Sami-mini', MAC Address '44:A3:BB:B4:36:D8', Assigned IP '192.168.0.104', and Lease Time '01:56:41'. The second client has ID 2, Client Name 'sam-E-s-Redmi-Note-1 4-Pro', MAC Address '52:84:2A:2A:24:6E', Assigned IP '192.168.0.101', and Lease Time '01:58:16'. A 'Refresh' button is located below the table. The bottom right corner of the interface features links for 'Support' and 'App'.

ID	Client Name	MAC Address	Assigned IP	Lease Time
1	Sami-mini	44:A3:BB:B4:36:D8	192.168.0.104	01:56:41
2	sam-E-s-Redmi-Note-1 4-Pro	52:84:2A:2A:24:6E	192.168.0.101	01:58:16

Step 3: Open Wireless MAC Filtering

The Wireless 5 GHz → Wireless MAC Filtering page was opened. Initially, MAC Filtering was disabled and no active filtering entry was present.



AC750 Wireless Dual Band Router
Model No. Archer C20

- Status
- Quick Setup
- Operation Mode
- Network
- Dual Band Selection
- Wireless 2.4GHz
- Wireless 5GHz**
 - Basic Settings
 - WPS
 - Wireless Security
 - Wireless MAC Filtering
 - Wireless Advanced
 - Wireless Statistics
- Guest Network
- DHCP
- Forwarding
- Security
- Parental Controls
- Access Control
- Advanced Routing
- Bandwidth Control
- IP & MAC Binding
- Dynamic DNS
- IPv6
- System Tools
- Logout

Wireless MAC Filtering

You can configure Wireless MAC Filtering which allows you to control wireless access on the network on this page:

Wireless MAC Filtering: Disabled

Filtering Rules

☒ Deny the stations specified by any enabled entries in the list to access.

☐ Allow the stations specified by any enabled entries in the list to access.

☐	MAC Address	Status	Host	Description	Edit
Add New Enable Selected Disable Selected Delete Selected					

[Support](#)
[App](#)

Step 4: Enable MAC Filtering and Select a Filtering Rule



AC750 Wireless Dual Band Router
Model No. Archer C20

- Status
- Quick Setup
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- Wireless 5GHz**
 - Basic Settings
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 - Wireless Security
 - Wireless MAC Filtering
 - Wireless Advanced
 - Wireless Statistics
- Guest Network
- DHCP
- Forwarding
- Security
- Parental Controls
- Access Control
- Advanced Routing
- Bandwidth Control
- IP & MAC Binding
- Dynamic DNS
- IPv6
- System Tools
- Logout

Wireless MAC Filtering

You can configure Wireless MAC Filtering which allows you to control wireless access on the network on this page.

Wireless MAC Filtering: Enabled

Filtering Rules

☒ Deny the stations specified by any enabled entries in the list to access.

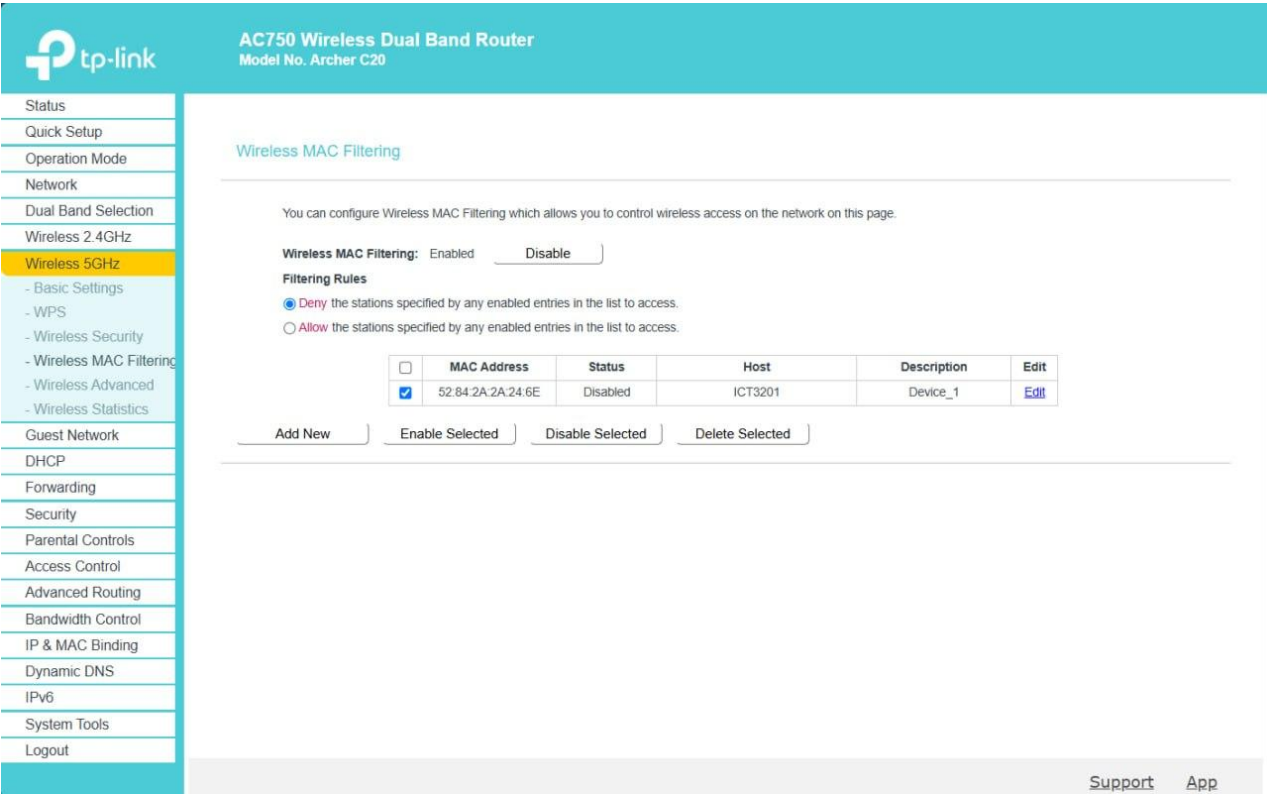
☐ Allow the stations specified by any enabled entries in the list to access.

☐	MAC Address	Status	Host	Description	Edit
Add New Enable Selected Disable Selected Delete Selected					

[Support](#)
[App](#)

Step 5: Add the Selected Device MAC Address

The MAC address of the selected client device was added to the filtering list. The entry was enabled so that the configured rule could be applied to that device.



Observation and Result:

The experiment successfully demonstrated the configuration of router-level wireless access control. The DHCP Clients List provided the MAC address and IP information of connected devices. Wireless MAC Filtering was then enabled on the 5 GHz network, and the selected device MAC address was added to the filtering table. Therefore, access to the wireless network can be controlled according to the configured Allow or Deny filtering policy.

Configuration Item	Observed Configuration	Result
Wireless Security	WPA2-PSK / AES	Wireless network protected with a password
DHCP Client Identification	Client name, MAC and IP displayed	Target device information identified
MAC Filtering	Enabled on 5 GHz network	Access-control policy activated
MAC Address Entry	Selected device added and enabled	Device access controlled by filtering rule

Discussion:

MAC Filtering provides an additional layer of administrative control by using the physical address of a device. The DHCP Clients List is useful because it allows the administrator to identify currently connected devices and obtain their MAC addresses without manually searching for them. After the MAC address is added to the filtering list, the selected Allow or Deny policy determines the device's ability to connect.

MAC Filtering should be considered an additional control rather than the only security mechanism. Strong wireless encryption and a secure password are still necessary because MAC addresses can be changed or imitated by advanced users. Combining WPA2/WPA3 security, strong passwords, and MAC-based access rules provides better overall user access control.

Conclusion:

The experiment successfully demonstrated router configuration and MAC Filtering for user access control. Wireless security was configured, connected clients were identified through the DHCP Clients List, and the target device MAC address was added to the Wireless MAC Filtering table. By enabling the appropriate Allow or Deny rule, the network administrator can control which devices are permitted to use the wireless network. This experiment shows how router configuration can be used to improve basic network security and manage user access effectively.